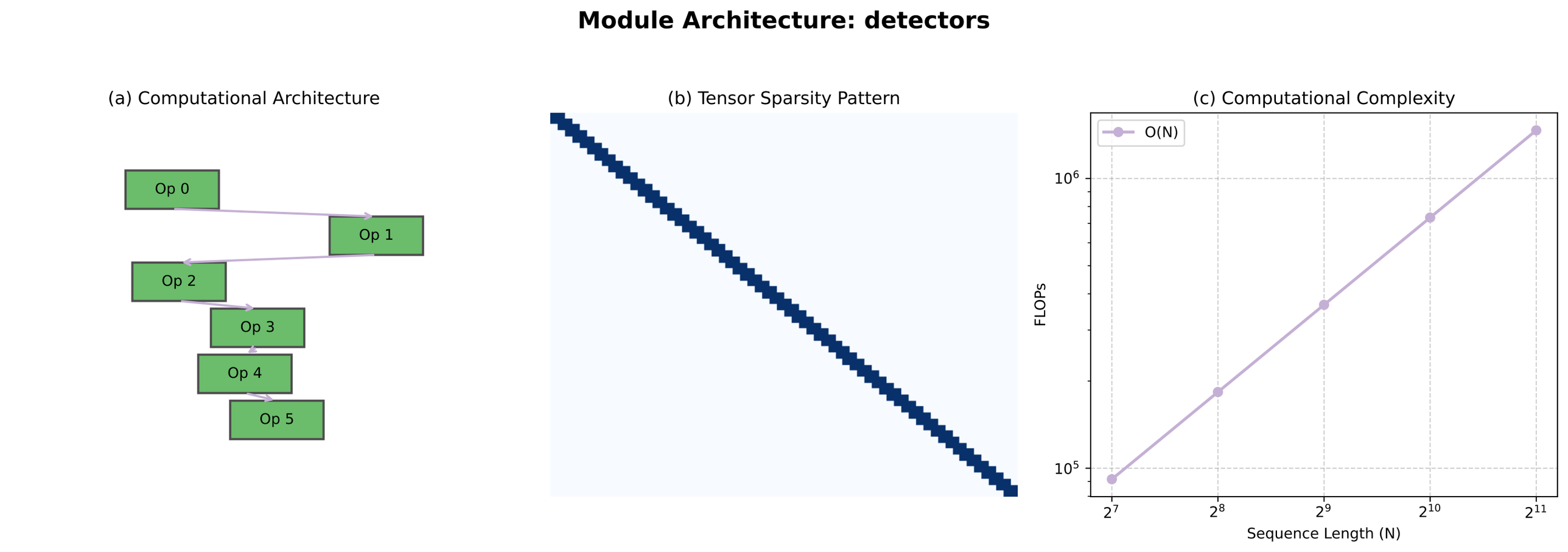Module Architecture: detectors